

Unitree Week 25

Presentation by Mario Ortiz

This Week

- Working on the Reference-Motion Observation Learning Algorithm
- Pipeline:
 - R1 Retargeted Motion → RL Dataset → PPO Training → Visual Evaluation

Retargeted R1 Motion Dataset + Library

- Using the previous set of Retargeted R1 motions, I created a dataset containing all of the motion files.
- The idea of this is to sample several motion files during training.
- Then, a motion Library loads the dataset and gives IsaacLab (the RL environment access to any access frames)
 - With `motion_id` → we access this files, `local_frame`, referent joint positions, velocities, root pose, root velocity, and phase through motion.

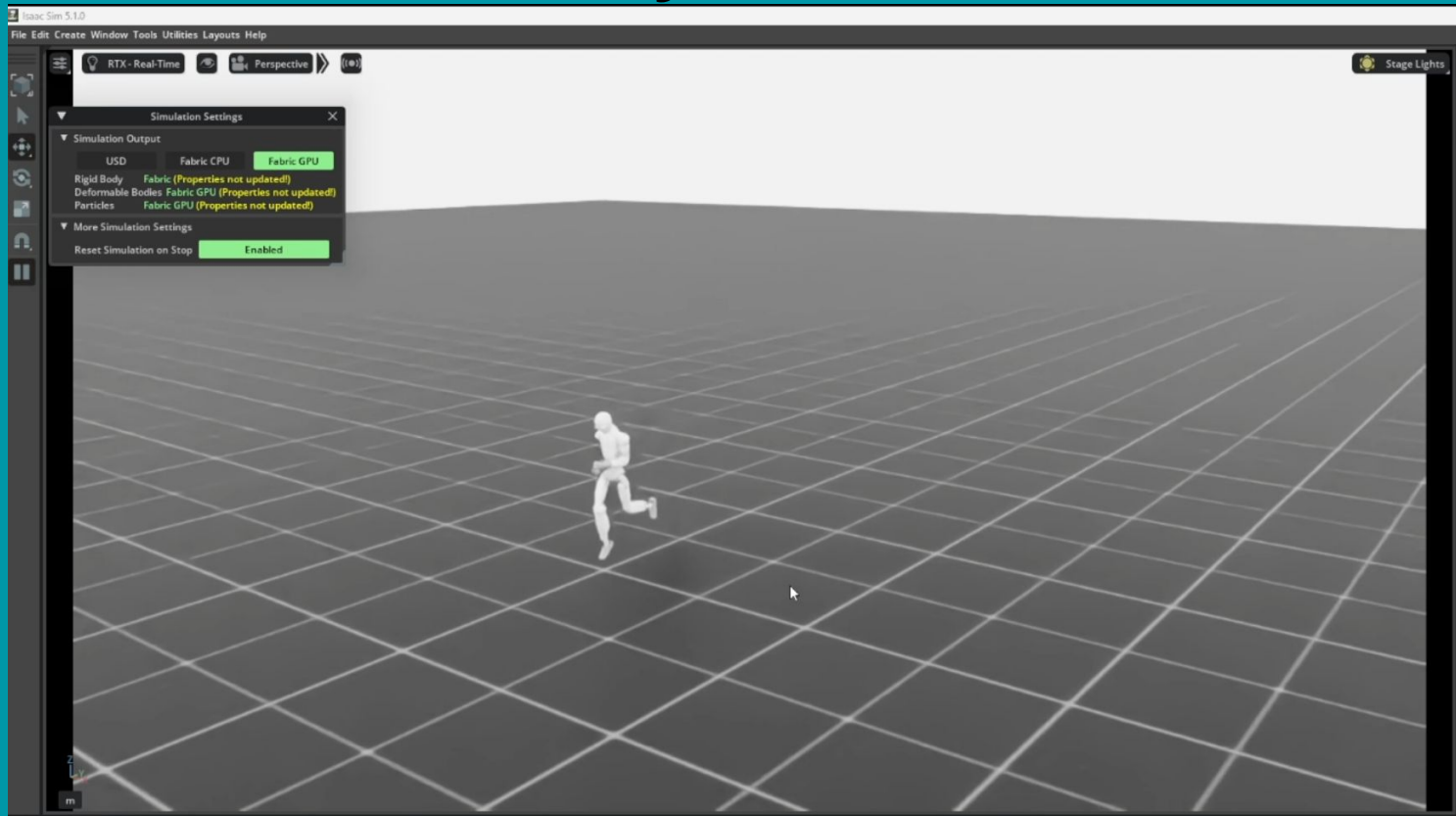
Important

- Previously, the retargeted motions developed from inverse kinematics were all shown on IsaacSim.
- Isaac Sim is the primary robotics simulator.
- Isaac Lab is NVIDIA's Reinforcement Learning API
 - The goal is to first implement motion files onto IsaacLab, ensuring they work.
 - Next, we train the R1 using these motion files with Reinforcement Learning.

Isaac Lab Reference Reset Test

- Next, a test was done to ensure that one reference frame can correctly place the R1 onto the desired pose.
 - Proves we can input any dataset frame and the R1 pose is initialized.
- We then test whether consecutive frames can be played after another (frame 1 → frame 2 → etc.)
 - Kinematic_playback: directly outputs the full root/joint state for every frame.

Kinematic Playback



IsaacLab Reinforcement Learning Environment

- Now we work on the Reinforcement Learning Environment, where we have the R1 following the reference motion physically.
- The algorithm works by:
 - Setting a policy action to a joint-position correction.
 - Setting joint target to the function $\text{reference_joint_pos} + \text{action_scale} * \text{policy_action}$.
 - The policy will learn corrections around the reference motion.

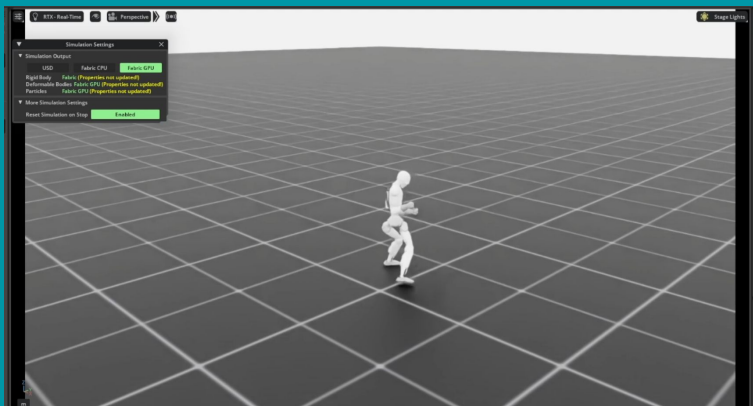
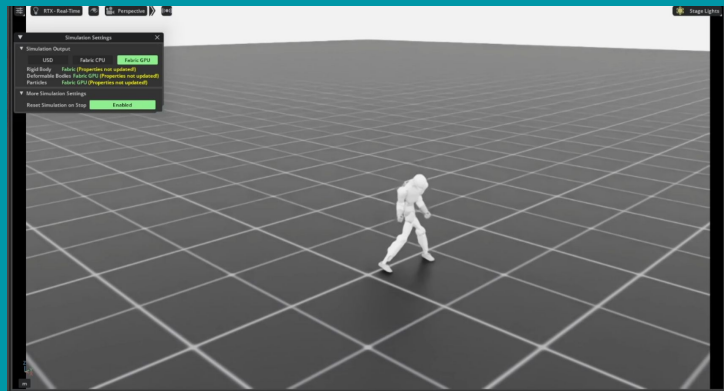
Proximal Policy Optimization (PPO) Training Script

- We run PPO onto the R1 motion-tracking environment.
- Initial Parameters:
 - Number of environments (num_envs) = 256
 - Number of iterations = 1000
- Results:
 - The training pipeline ran successfully, saving checkpoints.
 - The robot mostly learned to stay upright rather than imitating the exact motion of the motion file.

Proximal Policy Optimization (PPO) Training Script

iter=00001	steps=000032768	fps=05300	reward_mean= 4.954	done_count=0291	root_z_mean= 0.661	std= 0.351	pi_loss=-0.0199	vf_loss= 856.5129	entropy= 9.634	kl= 0.01283	clip= 0.186	ev= 0.003
iter=00002	steps=000040152	fps=05510	reward_mean= 4.962	done_count=0296	root_z_mean= 0.639	std= 0.351	pi_loss=-0.0172	vf_loss= 621.1711	entropy= 9.656	kl= 0.01045	clip= 0.149	ev= 0.038
iter=00003	steps=000065536	fps=05417	reward_mean= 4.963	done_count=0312	root_z_mean= 0.644	std= 0.351	pi_loss=-0.0163	vf_loss= 692.0567	entropy= 9.670	kl= 0.01112	clip= 0.158	ev= 0.238
iter=00004	steps=000081920	fps=05361	reward_mean= 4.952	done_count=0316	root_z_mean= 0.652	std= 0.351	pi_loss=-0.0188	vf_loss= 719.9186	entropy= 9.684	kl= 0.01475	clip= 0.203	ev= 0.411
iter=00005	steps=000098304	fps=05408	reward_mean= 4.958	done_count=0296	root_z_mean= 0.647	std= 0.352	pi_loss=-0.0142	vf_loss= 669.7536	entropy= 9.704	kl= 0.02035	clip= 0.274	ev= 0.531
iter=00006	steps=000114688	fps=05305	reward_mean= 4.929	done_count=0302	root_z_mean= 0.634	std= 0.352	pi_loss=-0.0253	vf_loss= 556.5576	entropy= 9.741	kl= 0.02098	clip= 0.272	ev= 0.653
iter=00007	steps=000131072	fps=05159	reward_mean= 4.933	done_count=0319	root_z_mean= 0.648	std= 0.353	pi_loss=-0.0229	vf_loss= 457.6158	entropy= 9.800	kl= 0.02064	clip= 0.276	ev= 0.766
iter=00008	steps=000147456	fps=05078	reward_mean= 4.966	done_count=0301	root_z_mean= 0.649	std= 0.354	pi_loss=-0.0236	vf_loss= 457.0374	entropy= 9.842	kl= 0.02266	clip= 0.303	ev= 0.788
iter=00009	steps=000163840	fps=05083	reward_mean= 4.958	done_count=0308	root_z_mean= 0.650	std= 0.354	pi_loss=-0.0256	vf_loss= 306.4804	entropy= 9.881	kl= 0.02316	clip= 0.300	ev= 0.861
iter=00010	steps=000180224	fps=05090	reward_mean= 4.943	done_count=0297	root_z_mean= 0.639	std= 0.354	pi_loss=-0.0205	vf_loss= 297.7867	entropy= 9.905	kl= 0.01994	clip= 0.268	ev= 0.858
iter=00011	steps=000196608	fps=05111	reward_mean= 4.957	done_count=0300	root_z_mean= 0.651	std= 0.355	pi_loss=-0.0239	vf_loss= 195.4017	entropy= 9.935	kl= 0.02025	clip= 0.272	ev= 0.909
iter=00012	steps=000212992	fps=05136	reward_mean= 4.936	done_count=0297	root_z_mean= 0.635	std= 0.355	pi_loss=-0.0214	vf_loss= 169.6908	entropy= 9.959	kl= 0.03452	clip= 0.378	ev= 0.923
iter=00013	steps=000229376	fps=05157	reward_mean= 4.946	done_count=0308	root_z_mean= 0.639	std= 0.355	pi_loss=-0.0237	vf_loss= 151.2317	entropy= 9.981	kl= 0.02553	clip= 0.320	ev= 0.930
iter=00014	steps=000245760	fps=05145	reward_mean= 4.954	done_count=0318	root_z_mean= 0.658	std= 0.356	pi_loss=-0.0268	vf_loss= 130.5225	entropy= 10.004	kl= 0.02627	clip= 0.324	ev= 0.940
iter=00015	steps=000262144	fps=05143	reward_mean= 4.937	done_count=0294	root_z_mean= 0.649	std= 0.356	pi_loss=-0.0250	vf_loss= 103.1242	entropy= 10.029	kl= 0.02533	clip= 0.317	ev= 0.952
iter=00016	steps=000278528	fps=05148	reward_mean= 4.938	done_count=0304	root_z_mean= 0.656	std= 0.356	pi_loss=-0.0250	vf_loss= 93.4984	entropy= 10.050	kl= 0.02526	clip= 0.305	ev= 0.956
iter=00017	steps=000294912	fps=05162	reward_mean= 4.938	done_count=0292	root_z_mean= 0.650	std= 0.356	pi_loss=-0.0248	vf_loss= 78.7432	entropy= 10.069	kl= 0.02801	clip= 0.337	ev= 0.961
iter=00018	steps=000311296	fps=05167	reward_mean= 4.935	done_count=0309	root_z_mean= 0.655	std= 0.357	pi_loss=-0.0203	vf_loss= 73.4560	entropy= 10.083	kl= 0.03074	clip= 0.360	ev= 0.966
iter=00019	steps=000327680	fps=05168	reward_mean= 4.945	done_count=0299	root_z_mean= 0.644	std= 0.357	pi_loss=-0.0229	vf_loss= 71.2676	entropy= 10.100	kl= 0.02236	clip= 0.283	ev= 0.967
iter=00020	steps=000344064	fps=05173	reward_mean= 4.932	done_count=0302	root_z_mean= 0.646	std= 0.357	pi_loss=-0.0247	vf_loss= 68.6344	entropy= 10.112	kl= 0.03235	clip= 0.368	ev= 0.968
iter=00021	steps=000360448	fps=05188	reward_mean= 4.921	done_count=0310	root_z_mean= 0.654	std= 0.357	pi_loss=-0.0259	vf_loss= 57.8958	entropy= 10.130	kl= 0.02341	clip= 0.303	ev= 0.973
iter=00022	steps=000376832	fps=05195	reward_mean= 4.936	done_count=0289	root_z_mean= 0.644	std= 0.358	pi_loss=-0.0253	vf_loss= 57.1671	entropy= 10.151	kl= 0.03011	clip= 0.341	ev= 0.972
iter=00023	steps=000393216	fps=05207	reward_mean= 4.928	done_count=0307	root_z_mean= 0.662	std= 0.358	pi_loss=-0.0246	vf_loss= 55.5082	entropy= 10.167	kl= 0.02582	clip= 0.322	ev= 0.975
iter=00024	steps=000409600	fps=05216	reward_mean= 4.955	done_count=0284	root_z_mean= 0.654	std= 0.358	pi_loss=-0.0276	vf_loss= 52.4098	entropy= 10.192	kl= 0.03445	clip= 0.371	ev= 0.976
iter=00025	steps=000425984	fps=05230	reward_mean= 4.932	done_count=0300	root_z_mean= 0.661	std= 0.359	pi_loss=-0.0212	vf_loss= 53.9686	entropy= 10.217	kl= 0.02427	clip= 0.301	ev= 0.977
iter=00026	steps=000442368	fps=05234	reward_mean= 4.953	done_count=0282	root_z_mean= 0.646	std= 0.359	pi_loss=-0.0269	vf_loss= 53.7251	entropy= 10.244	kl= 0.02882	clip= 0.332	ev= 0.977
iter=00027	steps=000458752	fps=05231	reward_mean= 4.949	done_count=0292	root_z_mean= 0.654	std= 0.359	pi_loss=-0.0270	vf_loss= 52.6854	entropy= 10.272	kl= 0.02482	clip= 0.308	ev= 0.978
iter=00028	steps=000475136	fps=05227	reward_mean= 4.958	done_count=0283	root_z_mean= 0.644	std= 0.360	pi_loss=-0.0254	vf_loss= 62.3688	entropy= 10.291	kl= 0.02595	clip= 0.320	ev= 0.975
iter=00029	steps=000491520	fps=05232	reward_mean= 4.951	done_count=0289	root_z_mean= 0.656	std= 0.360	pi_loss=-0.0228	vf_loss= 63.9043	entropy= 10.305	kl= 0.02200	clip= 0.297	ev= 0.975
iter=00030	steps=000507904	fps=05241	reward_mean= 4.944	done_count=0278	root_z_mean= 0.651	std= 0.360	pi_loss=-0.0228	vf_loss= 61.0048	entropy= 10.308	kl= 0.02396	clip= 0.299	ev= 0.975
iter=00031	steps=000524288	fps=05252	reward_mean= 4.925	done_count=0275	root_z_mean= 0.644	std= 0.360	pi_loss=-0.0292	vf_loss= 52.3339	entropy= 10.319	kl= 0.02924	clip= 0.338	ev= 0.979
iter=00032	steps=000540672	fps=05250	reward_mean= 4.937	done_count=0284	root_z_mean= 0.638	std= 0.360	pi_loss=-0.0224	vf_loss= 52.5051	entropy= 10.332	kl= 0.02623	clip= 0.324	ev= 0.979
iter=00033	steps=000557056	fps=05246	reward_mean= 4.946	done_count=0285	root_z_mean= 0.652	std= 0.360	pi_loss=-0.0251	vf_loss= 49.5961	entropy= 10.337	kl= 0.02427	clip= 0.318	ev= 0.979
iter=00034	steps=000573440	fps=05245	reward_mean= 4.945	done_count=0277	root_z_mean= 0.656	std= 0.360	pi_loss=-0.0245	vf_loss= 52.4071	entropy= 10.341	kl= 0.02475	clip= 0.311	ev= 0.979
iter=00035	steps=000589824	fps=05257	reward_mean= 4.957	done_count=0278	root_z_mean= 0.655	std= 0.360	pi_loss=-0.0280	vf_loss= 48.1226	entropy= 10.356	kl= 0.02193	clip= 0.290	ev= 0.980
iter=00036	steps=000606208	fps=05265	reward_mean= 4.956	done_count=0274	root_z_mean= 0.648	std= 0.361	pi_loss=-0.0284	vf_loss= 54.2061	entropy= 10.363	kl= 0.01800	clip= 0.246	ev= 0.979
iter=00037	steps=000622592	fps=05275	reward_mean= 4.960	done_count=0278	root_z_mean= 0.662	std= 0.361	pi_loss=-0.0255	vf_loss= 49.7827	entropy= 10.377	kl= 0.02542	clip= 0.312	ev= 0.981
iter=00038	steps=000638976	fps=05286	reward_mean= 4.954	done_count=0261	root_z_mean= 0.640	std= 0.361	pi_loss=-0.0224	vf_loss= 51.3212	entropy= 10.394	kl= 0.01842	clip= 0.251	ev= 0.980
iter=00039	steps=000655360	fps=05290	reward_mean= 4.978	done_count=0264	root_z_mean= 0.646	std= 0.361	pi_loss=-0.0240	vf_loss= 48.5154	entropy= 10.409	kl= 0.02305	clip= 0.294	ev= 0.981
iter=00040	steps=000671744	fps=05302	reward_mean= 4.954	done_count=0280	root_z_mean= 0.650	std= 0.361	pi_loss=-0.0270	vf_loss= 60.8073	entropy= 10.419	kl= 0.03180	clip= 0.364	ev= 0.981
iter=00041	steps=000688128	fps=05311	reward_mean= 4.967	done_count=0275	root_z_mean= 0.666	std= 0.361	pi_loss=-0.0223	vf_loss= 58.6360	entropy= 10.427	kl= 0.02644	clip= 0.328	ev= 0.979
iter=00042	steps=000704512	fps=05310	reward_mean= 4.963	done_count=0261	root_z_mean= 0.656	std= 0.362	pi_loss=-0.0164	vf_loss= 63.3261	entropy= 10.438	kl= 0.01847	clip= 0.259	ev= 0.979
iter=00043	steps=000720896	fps=05312	reward_mean= 4.973	done_count=0266	root_z_mean= 0.645	std= 0.361	pi_loss=-0.0290	vf_loss= 58.1649	entropy= 10.438	kl= 0.02318	clip= 0.294	ev= 0.978
iter=00044	steps=000737280	fps=05316	reward_mean= 4.975	done_count=0262	root_z_mean= 0.648	std= 0.361	pi_loss=-0.0235	vf_loss= 59.5844	entropy= 10.430	kl= 0.02823	clip= 0.338	ev= 0.979
iter=00045	steps=000753664	fps=05321	reward_mean= 4.964	done_count=0270	root_z_mean= 0.647	std= 0.362	pi_loss=-0.0289	vf_loss= 65.0718	entropy= 10.442	kl= 0.02226	clip= 0.288	ev= 0.977
iter=00046	steps=000770048	fps=05325	reward_mean= 4.982	done_count=0266	root_z_mean= 0.653	std= 0.362	pi_loss=-0.0277	vf_loss= 65.4929	entropy= 10.464	kl= 0.02596	clip= 0.318	ev= 0.977

Proximal Policy Optimization (PPO) Training Script



```
026-07-24T17:07:57Z [39,168ms] [Warning] [omni.fabric.plugin] getAttributeCount called on non-existent path /w
026-07-24T17:07:57Z [39,171ms] [Warning] [omni.fabric.plugin] getTypes called on non-existent path /World/R1/w
tep=00000 | frame=0000 | reward= 6.346 | done=0 | root_z= 0.820 | action_abs= 0.708 | actor_mean_abs= 0.924
026-07-24T17:08:05Z [47,633ms] [Warning] [carb] Client gpu.foundation.plugin has acquired [gpu::unstable::IMem
tep=00120 | frame=0156 | reward= 5.143 | done=0 | root_z= 0.636 | action_abs= 0.568 | actor_mean_abs= 0.675
tep=00240 | frame=0312 | reward= 5.473 | done=0 | root_z= 0.701 | action_abs= 0.791 | actor_mean_abs= 1.351
tep=00360 | frame=0468 | reward= 5.261 | done=0 | root_z= 0.680 | action_abs= 0.720 | actor_mean_abs= 0.954
tep=00480 | frame=0624 | reward= 5.026 | done=0 | root_z= 0.730 | action_abs= 0.516 | actor_mean_abs= 0.555

uto-reset 1: new_start_frame=703
tep=00000 | frame=0703 | reward= 6.363 | done=0 | root_z= 0.815 | action_abs= 0.714 | actor_mean_abs= 0.960

uto-reset 2: new_start_frame=885
tep=00000 | frame=0885 | reward= 6.350 | done=0 | root_z= 0.804 | action_abs= 0.718 | actor_mean_abs= 0.937
tep=00120 | frame=1041 | reward= 5.829 | done=0 | root_z= 0.734 | action_abs= 0.701 | actor_mean_abs= 0.919

uto-reset 3: new_start_frame=14
tep=00000 | frame=0014 | reward= 6.361 | done=0 | root_z= 0.819 | action_abs= 0.710 | actor_mean_abs= 0.937
CTerminate batch job (Y/N)?
```

Currently, the robots learn initially to stand up and move, though it doesn't at all mimic the task.

Exact Imitation for RL

- Currently it seems the Reinforcement Learning algorithms contain too much exploration, resulting in the robots not learning to follow the exact motion. (This makes sense as at the moment the goal was to simply obtain a running RL pipeline).
- As we currently have motion datasets, the best method to implement this to RL is to train the robot to follow the exact motion of each motion file frame-by-frame.
 - This will be done by rewarding the R1 for following the exact joint parameters (At frame t , match the exact dataset state at frame t)
 - Once it masters the motion of the frame, the next iteration will start from initial frame up until the next frame.
 - Example: (we train robot to complete frame 1 → next step we train robot to do frame 2 AFTER completing frame 1.